

Last Time

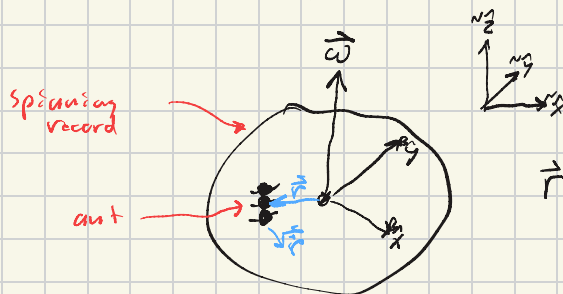
- Cart Pole
- Manipulator Equation
- Mass Matrix
- (started) Rotating Frames

Today

- Rotating Frames
- Rotation Matrix Kinematics
- Velocity Transport Theorem
- "Fictitious Forces"
- MEMS Gyro

Rotating Reference Frames

- Ant on a record toy problem:



$$\vec{r} = \begin{bmatrix} {}^N r_1 \\ {}^N r_2 \\ {}^N r_3 \end{bmatrix}^T \begin{bmatrix} {}^N x \\ {}^N y \\ {}^N z \end{bmatrix} = \begin{bmatrix} {}^B r_1 \\ {}^B r_2 \\ {}^B r_3 \end{bmatrix}^T \begin{bmatrix} {}^B x \\ {}^B y \\ {}^B z \end{bmatrix}$$

- ${}^N r$ and ${}^B r$ are related by a rotation:

$$\underbrace{\begin{bmatrix} {}^N x \\ {}^N y \\ {}^N z \end{bmatrix}}_I \underbrace{\begin{bmatrix} {}^N x \\ {}^N y \\ {}^N z \end{bmatrix}^T}_{{}^N R^B} \begin{bmatrix} {}^N r_1 \\ {}^N r_2 \\ {}^N r_3 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^N x \\ {}^N y \\ {}^N z \end{bmatrix}}_{} \underbrace{\begin{bmatrix} {}^B x \\ {}^B y \\ {}^B z \end{bmatrix}^T}_{{}^N R^B} \begin{bmatrix} {}^B r_1 \\ {}^B r_2 \\ {}^B r_3 \end{bmatrix}$$

$$\Rightarrow {}^N r = {}^N R^B {}^B r$$

function of t because of spinning

- Let's say ant is standing still on the record

$$\Rightarrow {}^B \dot{\mathbf{r}} = \mathbf{0}$$

- What's the ant's velocity in the N frame?

$$\begin{aligned} * \text{ From a little vector geometry: } {}^N \dot{\mathbf{r}} &= {}^N \boldsymbol{\omega} \times {}^N \mathbf{r} = (R^B \boldsymbol{\omega}) \times (R^B \mathbf{r}) \\ &= R({}^B \boldsymbol{\omega} \times {}^B \mathbf{r}) \end{aligned}$$

$$* \text{ From calculus: } {}^N \mathbf{r} = {}^N R(t) {}^B \mathbf{r} \Rightarrow {}^N \dot{\mathbf{r}} = \dot{R} {}^B \mathbf{r} + R \cancel{{}^B \dot{\mathbf{r}}} \quad \text{no}$$

- Combine these:

$${}^N \dot{\mathbf{r}} = \dot{R} {}^B \mathbf{r} = {}^N \dot{R} ({}^B \boldsymbol{\omega} \times {}^B \mathbf{r})$$

Introduce "Cross-Product Matrix" / "Hat Map"

$$\boldsymbol{\omega} \times \mathbf{x} = \begin{bmatrix} \omega_2 x_3 - \omega_3 x_2 \\ \omega_3 x_1 - \omega_1 x_3 \\ \omega_1 x_2 - \omega_2 x_1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}}_{\hat{\boldsymbol{\omega}} / \boldsymbol{\omega}^\times / [\boldsymbol{\omega}]^\times / S(\boldsymbol{\omega})} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- Note, because $\boldsymbol{\omega} \times \mathbf{x} = -\mathbf{x} \times \boldsymbol{\omega}$, we have the following:

$$* S(\boldsymbol{\omega}) = -S(\boldsymbol{\omega})^T \quad (\text{"skew symmetric"})$$

$$* S(\boldsymbol{\omega}) \mathbf{x} = -S(\mathbf{x}) \boldsymbol{\omega}$$

- Plug in hat map for $\boldsymbol{\omega} \times$:

$$\Rightarrow {}^N \dot{\mathbf{r}} = \dot{R} {}^B \mathbf{r} = R S({}^B \boldsymbol{\omega}) {}^B \mathbf{r}$$

- must be true for any ${}^B \mathbf{r}$

$$\Rightarrow \boxed{{}^N \dot{R}^B = {}^N R^B S(\omega)} \quad \leftarrow \text{can use this to integrate gyro into } R_{GB}$$

* "Velocity Transport Theorem":

- assuming ${}^B \dot{r} \neq 0$

$$\Rightarrow {}^N r = R_{GB} {}^B r \Rightarrow {}^N \dot{r} = \dot{R} {}^B r + R {}^B \dot{r}$$

$$= \dot{R} {}^B r + R S(\omega) {}^B r$$

$$\Rightarrow \boxed{{}^N \dot{r} = R ({}^B \dot{r} + {}^B \omega \times {}^B r)}$$

motion in
B frame

due to motion of frame

Newton's 2nd Law for the Ant

- Our ant is not in an inertial frame $\Rightarrow F=ma$ doesn't hold

- Let's derive an equivalent of Newton's 2nd law for the ant by writing down the Lagrangian of a point mass:

$$T = \frac{1}{2} m {}^N \dot{r}^T {}^N \dot{r} = \frac{1}{2} m ({}^B \dot{r} + \omega \times {}^B r)^T \underbrace{R^T R}_I ({}^B \dot{r} + \omega \times {}^B r)$$

$$= m \left[\frac{1}{2} {}^B \dot{r}^T {}^B \dot{r} + {}^B \dot{r}^T (\omega \times {}^B r) + \frac{1}{2} (\omega \times {}^B r)^T (\omega \times {}^B r) \right]$$

$$U = 0 \Rightarrow L = T$$

$$\frac{\partial L}{\partial \dot{r}} = m \dot{r} + m(\omega \times r)$$

$$\frac{\partial L}{\partial r} = \frac{\partial}{\partial r} [m \dot{r}^T (\omega \times r)] + \frac{\partial}{\partial r} \left[\frac{1}{2} m (\omega \times r)^T (\omega \times r) \right]$$

$$\rightarrow = \frac{\partial}{\partial r} [m \dot{r}^T S(\omega) r] = m \dot{r}^T S(\omega) = m(\dot{r} \times \omega) = -m \omega \times \dot{r}$$

$$\rightarrow = \frac{\partial}{\partial r} \left[\frac{1}{2} m \dot{r}^T S(\omega)^T S(\omega) r \right] = m \dot{r}^T S(\omega)^T S(\omega) = -m \omega \times \omega \times r$$

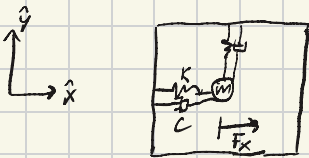
$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = m \ddot{r} + m \omega \times \dot{r} + m \dot{\omega} \times r$$

"Fictitious Forces"

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = m \ddot{r} + \underbrace{2m \omega \times \dot{r}}_{\text{"Coriolis"}} + \underbrace{m \dot{\omega} \times r}_{\text{"Euler / Transverse"}} + \underbrace{m \omega \times \omega \times r}_{\text{"Centrifugal"}} = 0$$

MEMS Gyro

- How do we measure ω e.g. on a drone (or your phone)?
- We'll look at a simplified picture of a single-axis sensor:



- 2D damped SHO

- Drive force applied in x-direction at resonant frequency $\omega_0 = \sqrt{\frac{k}{m}}$

$$F_x = a \sin(\omega_0 t)$$

- Assume rotation is only along z-axis $\Rightarrow {}^B\omega = \begin{bmatrix} 0 \\ 0 \\ \omega_z \end{bmatrix}$
- Assume isotropic stiffness + damping

$$\Rightarrow U = \frac{1}{2} k r^T r, \quad Q_{\text{damping}} = -c \dot{r}$$

$$- T = \frac{1}{2} m \dot{r}^T \dot{r} = \frac{1}{2} m (\dot{r} + \omega \times r)^T R^T R (\dot{r} + \omega \times r)$$

Same as before

- Only new things are spring, damping, and drive:

$$\Rightarrow m \ddot{r} + 2m\omega \times \dot{r} + m\dot{\omega} \times r + m\omega \times \omega \times r + k r + c \dot{r} = \begin{bmatrix} a \sin(\omega_0 t) \\ 0 \\ 0 \end{bmatrix}$$

* Simplifying assumptions:

- Ignore z-axis dynamics
- Assume $\omega_z \ll \omega_0 = \sqrt{\frac{m}{m}}$ ($\omega_0 \sim 20 \text{ kHz}$ for MEMS gyros)
- Assume $\dot{\omega}_z$ small

$$\Rightarrow \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix} + 2 \begin{bmatrix} 0 & -\omega_z \\ \omega_z & 0 \end{bmatrix} \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \end{bmatrix} + \frac{k}{m} \begin{bmatrix} r_x \\ r_y \end{bmatrix} + \frac{c}{m} \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \end{bmatrix} = \begin{bmatrix} a \sin(\omega_0 t) \\ 0 \end{bmatrix}$$

$$\Rightarrow \ddot{r}_x + \frac{k}{m} r_x + \frac{c}{m} \dot{r}_x = \underbrace{a \sin(\omega_0 t)}_{\text{drive } F_x} + \underbrace{2\omega_z \dot{r}_y}_{\text{small compared to drive}}$$

$$\underbrace{\ddot{r}_y + \frac{k}{m} r_y + \frac{c}{m} \dot{r}_y}_{\text{damped SHO}} = \underbrace{-2\omega_z \dot{r}_x}_{\text{drive proportional to } \omega_z}$$